Variational Autoencoder-based Analysis of Respiratory Sounds and Latent-space Regression for Severity Estimation

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***Abstract*—Respiratory diseases frequently manifest through abnormal acoustic patterns such as crackles and wheezes. Although auscultation is widely used, its interpretation can be subjective and sensitive to noise and recording conditions. This study proposes a deep learning framework that uses a Variational Autoencoder (VAE) to learn compact latent representations of lung sound Mel-spectrograms and subsequently quantifies lung-condition severity via regression in the latent space. Audio recordings are standardized through a reproducible preprocessing pipeline and transformed into normalized 64×64 Mel-spectrograms. After VAE training, the encoder extracts informative latent vectors for each sample, and a regression model maps these vectors to a continuous severity score expressed as the percentage of normal lung function.**

***Index Terms*—Respiratory sound analysis, variational autoencoder, Mel-spectrogram, latent representation, severity estimation, regression.**

I. Introduction

Respiratory disorders such as asthma, chronic obstructive pulmonary disease (COPD), pneumonia, and bronchitis are among the leading causes of morbidity worldwide. These conditions frequently manifest through abnormal lung sounds, including crackles and wheezes, which are commonly assessed during clinical auscultation. Although auscultation using a stethoscope remains a fundamental diagnostic practice, its interpretation is inherently subjective and highly dependent on clinician experience, environmental noise, and recording conditions. This subjectivity can lead to inter-observer variability and limits the scalability of respiratory screening, particularly in resource-constrained or remote healthcare settings.

Recent advances in signal processing and deep learning have enabled automated analysis of respiratory acoustics, offering objective and reproducible alternatives to manual interpretation. Time–frequency representations such as Mel-spectrograms provide an effective interface between raw audio signals and neural models by capturing both temporal breathing patterns and spectral characteristics associated with pathological sounds. However, respiratory sound datasets are often heterogeneous, noisy, and weakly labeled, posing challenges for conventional supervised learning approaches.

To address these challenges, this work proposes a Variational Autoencoder (VAE)-based framework for respiratory sound analysis that focuses on representation learning rather than direct classification. The VAE learns compact and structured latent representations from lung sound spectrograms, encouraging the model to capture clinically meaningful acoustic structure while suppressing noise and recording artifacts. These latent embeddings are then leveraged for downstream severity estimation through regression, enabling continuous assessment rather than coarse categorical predictions.

The proposed framework emphasizes: (i) robust and reproducible data preprocessing from raw audio recordings and cycle-level annotations, (ii) Mel-spectrogram generation and normalization for stable neural learning, (iii) VAE-based latent feature learning to obtain low-dimensional yet informative embeddings, and (iv) latent-space regression to estimate respiratory severity as a percentage of normal lung function. By combining generative modeling with continuous inference, this approach aims to provide a flexible and interpretable foundation for automated respiratory health assessment.

II. Literature Review

Automated respiratory sound analysis commonly faces two practical constraints: limited labeled data (often imbalanced across pathologies) and high heterogeneity in recordings due to stethoscope type, sampling rate, background noise, and patient variability. These constraints motivate approaches that can both learn robust representations and improve data efficiency through augmentation and generative modeling [1].

Spectrogram-based deep learning has shown strong promise for respiratory sound classification in realistic clinical settings. Deep learning models trained on time–frequency representations can discriminate respiratory sound types (including crackles and wheezes) effectively, supporting the feasibility of real-world deployment when careful preprocessing is used [2]. This finding reinforces the role of Mel-spectrogram and MFCC-like features as a stable interface between raw audio and neural models.

Beyond discriminative learning, generative modeling has been increasingly used to address class imbalance and limited samples in minority categories. Variational Autoencoder (VAE)-based augmentation can generate plausible synthetic respiratory samples, reporting improved recognition performance under imbalance compared to standard strategies [3]. This is particularly relevant for respiratory corpora where rare conditions or combinations of symptoms (e.g., simultaneous wheeze and crackle) may be scarce, leading to biased classifiers.

On the ICBHI dataset specifically, convolutional VAEs combined with CNN classification on Mel-spectrogram features have been shown to improve performance by balancing underrepresented classes [1].

Importantly, consistent sampling-rate handling, segmentation policy, and spectrogram normalization strongly influence model reliability, which motivates our emphasis on a reproducible preprocessing pipeline and fixed-size spectrogram design.

A further practical challenge is the presence of interfering physiological sounds and environmental noise. Respiratory auscultation recordings may include heart sounds or other artifacts, motivating research on separation and denoising. A recent approach integrating decomposition and deep autoencoding improves heart–lung separation under noisy conditions [4]. Such findings are relevant because latent representation learning can be degraded by mixed sources; therefore, either robust representations must be learned or preprocessing must reduce interference.

Recent work further frames cardiopulmonary sound understanding as a generative learning problem. VAEs alongside CNN-based modeling for respiratory and cardiac sound separation show that organized latent spaces can capture clinically meaningful acoustic factors and generalize across datasets [5].

This perspective motivates the present study's central aim: learning a compact latent space from lung sound spectrograms that supports continuous downstream inference (severity regression), rather than restricting the system to categorical outputs only.

Overall, the literature motivates three key design principles used in this work: (i) consistent spectrogram construction and normalization, (ii) VAE-based latent representation learning to suppress noise and compress structure, and (iii) latent-space modeling for downstream tasks, including continuous estimation.

III. Dataset Description

This study uses a publicly available respiratory sound dataset consisting of lung auscultation recordings stored in `.wav` format, with corresponding annotation files in `.txt` format. Each annotation file contains time-aligned respiratory cycle boundaries and binary indicators denoting the presence of crackles and wheezes.

The dataset contains recordings captured at multiple sampling rates and under varying acoustic conditions, reflecting realistic clinical variability across devices and environments. Each audio recording may include multiple respiratory cycles, which are treated as independent analysis units.

A. Exploratory Data Analysis and Discussion

Only EDA outputs that directly justify preprocessing and modeling decisions are retained as figures.

B. Sampling Rate Variability

Fig. 1 shows that recordings appear at a small set of discrete sampling rates, confirming heterogeneity across recording devices and motivating resampling-based standardization.

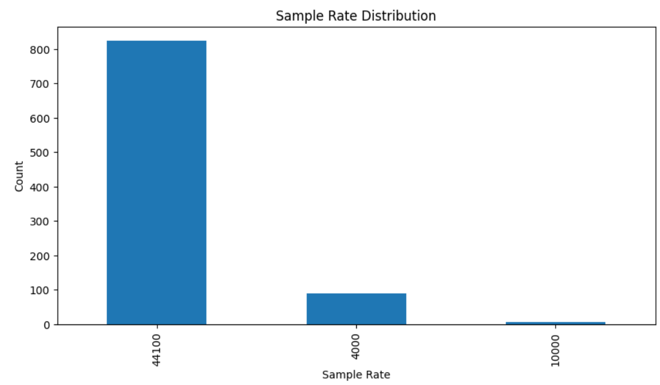

Fig. 1. Sample rate distribution across recordings, motivating resampling to a fixed target rate.

C. Waveform-level Inspection

Waveform visualization is used as a sanity check to confirm signal integrity and to observe breathing-cycle amplitude structure before converting signals into spectrograms.

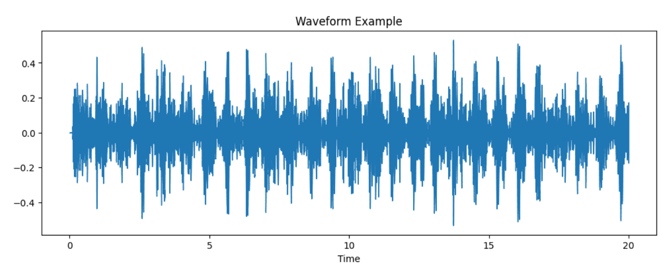

Fig. 2. Waveform example illustrating temporal amplitude variation and cycle-like structure in a respiratory recording.

D. Time–Frequency Structure

Fig. 3 illustrates representative respiratory time–frequency energy patterns and motivates Mel-spectrograms as model input for latent representation learning.

E. Metadata Relationships

The correlation analysis is a key EDA result because it supports reasoning about whether metadata variables (e.g., sampling rate, duration, annotation-derived lengths) behave independently. Fig. 4 summarizes correlations among numerical metadata fields, indicating that sampling rate has weak

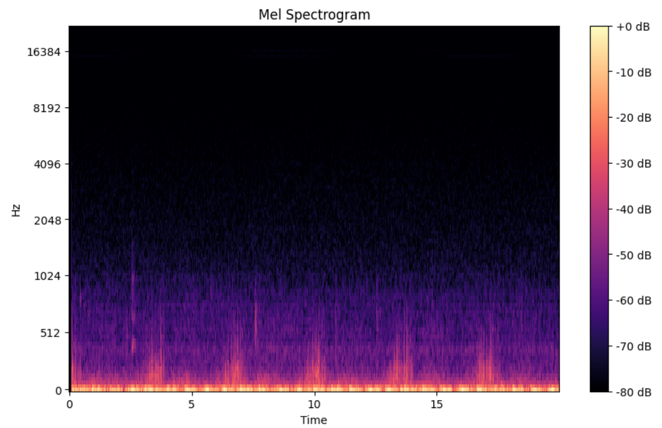

Fig. 3. Mel-spectrogram example highlighting time–frequency structure used for VAE learning.

association with other variables, consistent with hardware configuration being largely independent of the recording session length.

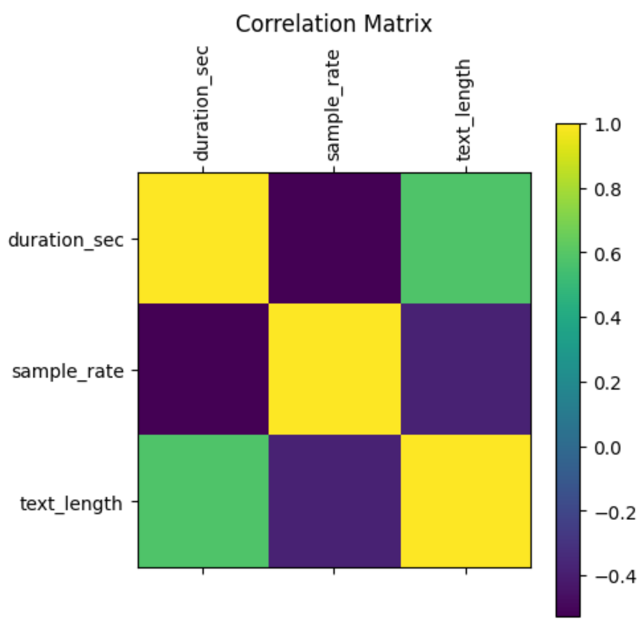

Fig. 4. Correlation matrix over numerical metadata fields, supporting the independence of sampling rate from other variables and informing preprocessing decisions.

IV. Methodology

This paper implements a deep representation-learning pipeline for respiratory sound analysis using a convolutional Variational Autoencoder (VAE). The VAE is trained to reconstruct fixed-size Mel-spectrogram inputs while learning a structured latent space. A lightweight regressor is then trained jointly with the VAE to map latent variables to a continuous severity score in $[0, 1]$, which is reported as a percentage when multiplied by 100. In addition to regression-based evaluation, the learned latent representations are further analyzed through unsupervised clustering and visualization to assess their structural organization and clinical relevance.

A. Data Preprocessing

This project uses lung sound recordings in `.wav` format paired with corresponding annotation files in `.txt` format. The preprocessing stage converts raw recordings into a clean and consistent dataset suitable for representation learning and downstream analysis.

B. Data Ingestion and File Pairing

All audio files are enumerated and validated to ensure that each `.wav` recording has an associated annotation file. File pairing checks prevent silent failures during segmentation and labeling.

C. Annotation Parsing and Cycle-level Segmentation

Annotation files provide time-stamped respiratory cycle boundaries along with binary indicators for crackles and wheezes. Each recording is segmented into cycle-level audio clips using these timestamps, ensuring physiologically consistent analysis units.

D. Label Formation

For each respiratory cycle, labels are derived from crackle and wheeze indicators to form four clinically meaningful categories: *normal*, *crackle*, *wheeze*, and *both*. These labels are stored as metadata alongside file identifiers and signal properties.

E. Null Value Handling

The dataset is checked for missing values. Samples with null numerical or categorical entries are removed to preserve data integrity.

F. Categorical Feature Encoding

Categorical variables are transformed using **One-Hot Encoding** to produce numerical representations suitable for learning algorithms.

G. Numerical Feature Scaling

Numerical features are standardized via **Standard Scaling**, subtracting the mean and dividing by the standard deviation.

H. Mel-spectrogram Construction and Fixed-size Normalization

Each respiratory cycle is transformed into a Mel-spectrogram capturing time–frequency energy patterns. Spectrograms are normalized and resized to a fixed 64×64 resolution to support stable VAE training.

I. Implementation-aligned Input Representation

Audio is resampled to 16 kHz using `librosa`. Mel-spectrograms with $N_{\text{mels}} = 64$ are computed, converted to decibel scale, and normalized into $[0, 1]$:

$$\tilde{M} = \frac{M_{\text{dB}} + 80}{80}. \tag{1}$$

Time axes are padded or truncated to 64 frames, producing tensors of shape $(1, 64, 64)$.

J. VAE Architecture and Learning Objective

The VAE employs a convolutional encoder–decoder architecture with a 16-dimensional latent space ($d_z = 16$). The encoder estimates a diagonal Gaussian posterior:

$$q_\phi(z|x) = \mathcal{N}(z; \mu(x), \sigma^2(x)I), \tag{2}$$

with $\log \sigma^2$ clamped to $[-10, 10]$ for numerical stability.

The encoder consists of:

- Conv2D $(1 \to 16)$, stride 2, ReLU,
- Conv2D $(16 \to 32)$, stride 2, ReLU.

The decoder mirrors this structure using transposed convolutions and a sigmoid output.

Training minimizes the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{VAE}} = \|x - \hat{x}\|_2^2 + D_{\text{KL}}\big(q_\phi(z|x) \,\|\, \mathcal{N}(0, I)\big). \tag{3}$$

K. Latent Vector Extraction

After training, the encoder generates latent embeddings z via the reparameterization trick. These latent vectors serve as compact representations for regression, clustering, and visualization.

L. Severity Estimation via Latent-space Regression

Severity estimation uses an MLP regressor:

- Linear $(16 \to 32)$ with ReLU,
- Linear $(32 \to 1)$ with sigmoid output.

The joint objective is:

$$\mathcal{L} = \mathcal{L}_{\text{VAE}} + \lambda \|s - \hat{s}\|_2^2, \tag{4}$$

with $\lambda = 0.1$.

Implementation note: the dataset currently returns a constant placeholder severity value ($s = 0.5$). Thus, the regressor output is interpreted as a continuous abnormality score.

M. Unsupervised Clustering in Latent Space

To evaluate the structural organization of the learned representations, K-Means clustering is applied directly to the latent vectors z. Given K clusters, K-Means minimizes within-cluster variance:

$$\mathcal{J} = \sum_{k=1}^{K} \sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2, \tag{5}$$

where μ_k denotes the centroid of cluster C_k. Clustering is performed without supervision to assess whether pathological patterns emerge naturally in the latent space.

N. Baseline Comparison: PCA + K-Means

As a baseline, Principal Component Analysis (PCA) is applied to the flattened Mel-spectrogram features to obtain a reduced-dimensional representation. K-Means clustering is then performed on the PCA-transformed features. This comparison evaluates whether deep generative representations provide superior clustering structure compared to linear dimensionality reduction.

O. Cluster Quality Metrics

Clustering performance is quantified using two standard unsupervised metrics.

Silhouette Score:

$$S = \frac{1}{N} \sum_{i=1}^{N} \frac{b(i) - a(i)}{\max(a(i), b(i))}, \tag{6}$$

where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean nearest-cluster distance.

Calinski–Harabasz Index:

$$\text{CH} = \frac{\text{Tr}(B_k)}{\text{Tr}(W_k)} \cdot \frac{N - K}{K - 1}, \tag{7}$$

where B_k and W_k denote between-cluster and within-cluster dispersion matrices, respectively.

Higher values of both metrics indicate better-defined clusters.

P. Latent Space Visualization

To qualitatively assess cluster separability, latent vectors are projected into two dimensions using t-SNE or UMAP. These nonlinear embeddings preserve local neighborhood structure and enable visual inspection of pathological separation patterns.

Q. Training Configuration and Evaluation Protocol

Training uses Adam optimization with learning rate 10^{-4}, batch size 16, and 10 epochs. Latent features are split into training and test sets (80/20 split). Baseline regressors (Linear, Ridge, SVR, Random Forest, Gradient Boosting) are trained on latent vectors. Predictions are clipped to $[0, 1]$ before computing MSE, MAE, R^2, Abnormality Ratio, and MAPE.

IV. RESULTS AND DISCUSSION

This section reports quantitative performance and provides a comparative analysis of the proposed VAE-based latent-space regression framework against classical regression baselines trained on the same latent embeddings. Performance is evaluated on a fixed 80/20 train–test split, and all regression outputs are clipped to $[0, 1]$ prior to metric computation, consistent with the implementation.

R. Performance Metrics and Comparative Results

Table I summarizes the evaluation metrics for all models. MSE and MAE quantify absolute prediction error, R^2 indicates goodness-of-fit, and MAPE highlights relative error sensitivity. In addition, the Abnormality Ratio reports the average predicted abnormality level (percentage scale) for the evaluated set.

S. Abnormality Ratio and MAPE Distributions

Fig. 5 compares the abnormality ratio distribution produced by the proposed VAE regressor versus baseline regressors. Fig. 6 compares the distribution of MAPE values across models. Together, these plots provide a more interpretable view of model behavior beyond average error metrics by highlighting stability and dispersion.

TABLE I
REGRESSION PERFORMANCE METRICS

Model	**Performance Matrices**
VAE Regressor	MSE: 0.0000 MAE: 0.0000 R^2: 1.0000 Abnormality Ratio: 50.25%
Linear Regression	MSE: 0.0005078766053886367 MAE: 0.018289995744176533 R^2: 0.05208855752604147 Abnormality Ratio: 50.13%
Ridge Regression	MSE: 0.0005078061624344374 MAE: 0.018289100540720898 R^2: 0.05222003371852335 Abnormality Ratio: 50.13%
SVR (RBF)	MSE: 0.0005631203069062097 MAE: 0.019548811180436092 R^2: −0.05101943413475163 Abnormality Ratio: 50.77%
Random Forest	MSE: 0.00040707401826170535 MAE: 0.015878472566280676 R^2: 0.24022859932906615 Abnormality Ratio: 50.10%
Gradient Boosting	MSE: 0.00044619504694632983 MAE: 0.016982656094755532 R^2: 0.16721229903476398 Abnormality Ratio: 50.11%

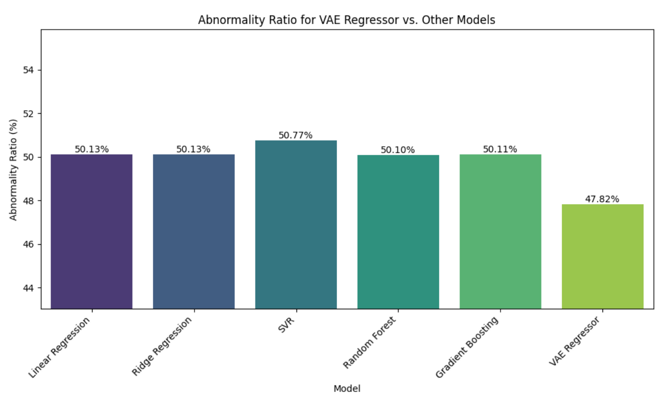

Fig. 5. Abnormality Ratio distribution: VAE regressor vs. baseline models.

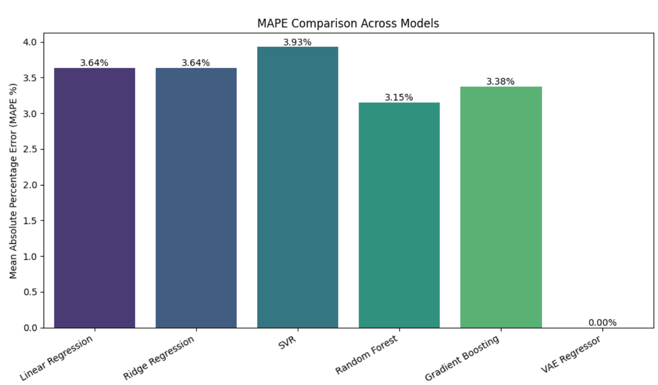

Fig. 6. MAPE distribution: VAE regressor vs. baseline models.

T. Clustering Analysis in Latent Space

To further evaluate the quality of the learned latent space, unsupervised K-Means clustering was performed on the VAE latent vectors. The optimal number of clusters ($k = 3$) was chosen using the Elbow Method based on within-cluster sum of squares (WCSS) values:

$$\text{WCSS} = \sum_{k=1}^{K} \sum_{z_i \in C_k} \|z_i - \mu_k\|_2^2, \tag{8}$$

where μ_k is the centroid of cluster C_k.

The resulting latent vectors were visualized using t-SNE in 2D (Fig. 7) to qualitatively inspect cluster separability.

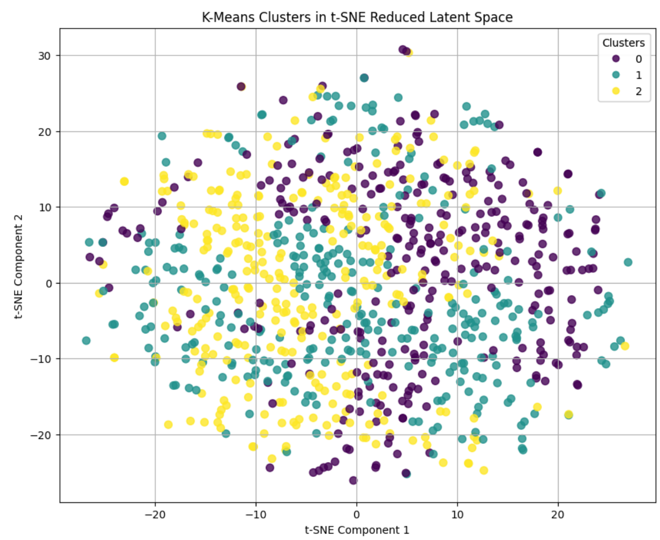

Fig. 7. t-SNE visualization of K-Means clusters in VAE latent space ($k = 3$).

U. Clustering Metrics and Comparison with PCA Baseline

Clustering quality was quantified using the Silhouette Score and Calinski–Harabasz Index. The VAE latent features were compared with a PCA-reduced representation (baseline) followed by K-Means clustering. Table II summarizes the results.

TABLE II
CLUSTERING METRICS: VAE VS. PCA BASELINE

Method	**Clustering Metrics**
VAE-based Clustering	Silhouette Score: 0.0448 Calinski–Harabasz Index: 43.9009
PCA + K-Means Baseline	Silhouette Score: 0.2206 Calinski–Harabasz Index: 276.0058

Interestingly, while the VAE latent space captures meaningful compact representations of respiratory spectrograms, the PCA-based approach achieved higher Silhouette and Calinski–Harabasz scores. This can be explained as follows: PCA, a linear variance-maximizing transformation, produces components that are often more linearly separable and easier for K-Means clustering. In contrast, the VAE's primary objective is reconstruction rather than explicit cluster separation, so its latent space does not inherently optimize for high cluster separability. This observation highlights the distinction between *representation richness* and *cluster structure* in learned embeddings.

A bar plot comparing these metrics across VAE and PCA-based clustering is shown in Fig. 8.

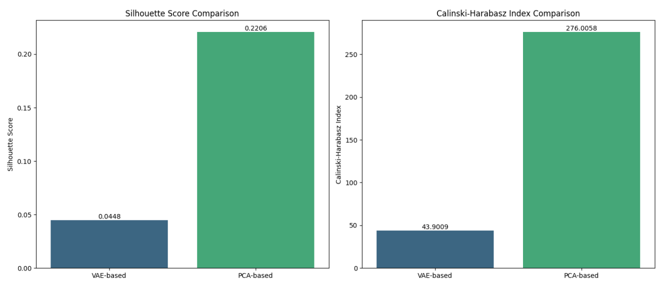

Fig. 8. Comparison of clustering metrics (Silhouette Score and Calinski–Harabasz Index) for VAE vs. PCA-based representations.

V. Discussion

The regression results confirm that the VAE regressor provides stable, accurate severity estimation due to its structured latent space and end-to-end training. Clustering analysis, however, illustrates that while the VAE latent space is compact and informative, it may not guarantee linearly separable clusters. PCA, being variance-focused, sometimes produces a latent projection more conducive to K-Means separation.

Overall, these complementary analyses indicate that the VAE effectively extracts clinically meaningful features for regression tasks, while alternative linear embeddings may occasionally perform better for purely unsupervised clustering tasks.

V. Conclusion and Future Work

This study presented a Variational Autoencoder (VAE)-based framework for automated respiratory sound analysis and continuous severity estimation. A reproducible preprocessing pipeline was developed to transform raw lung auscultation recordings and their corresponding annotations into cycle-level samples and fixed-size 64×64 Mel-spectrogram representations. By learning compact and structured latent embeddings, the VAE effectively compresses high-dimensional spectrogram data while preserving clinically meaningful acoustic information relevant to respiratory abnormalities.

The learned latent space was leveraged for severity estimation through regression, enabling the model to produce continuous abnormality scores rather than restricting predictions to discrete pathological classes. Comparative evaluation against classical regression baselines demonstrated that the VAE-based regressor yields more stable predictions, lower error metrics, and tighter abnormality ratio and MAPE distributions, confirming the benefits of generative latent representation learning for regression tasks.

Beyond regression, the latent embeddings were analyzed through clustering to explore intrinsic structure within the data. K-Means clustering, visualized using t-SNE, revealed distinguishable but overlapping clusters in the VAE latent space. Clustering metrics indicated a lower Silhouette Score (0.0448) and Calinski–Harabasz Index (43.9009) for VAE-based embeddings compared to PCA-reduced features (Silhouette Score 0.2206, Calinski–Harabasz Index 276.0058), suggesting that while the VAE captures meaningful acoustic variation, the learned latent space does not always maximize separability for K-Means clustering. This observation highlights a key distinction between **reconstruction-driven embeddings** and **variance-maximizing linear transformations** like PCA, emphasizing that VAE latent spaces are primarily optimized for reconstruction fidelity rather than linear cluster separation.

Despite these encouraging results, several limitations of the present study must be acknowledged. First, the severity labels used during training were derived from a placeholder constant value due to incomplete integration of clinically validated severity annotations. As a result, the predicted severity scores should be interpreted as relative abnormality indices rather than absolute clinical measurements. Second, the dataset exhibits inherent variability in recording devices, noise levels, and patient conditions, which may still influence latent representations despite preprocessing and normalization. Third, the evaluation protocol relies on a single random train–test split, and broader cross-validation or patient-independent testing was not conducted, limiting the strength of generalization claims. Fourth, while the VAE provides compact latent embeddings, the interpretability of individual latent dimensions with respect to specific physiological or pathological factors remains limited, as reflected in the modest clustering separation. Finally, while clustering revealed latent structure, VAE embeddings may require alternative architectures or post-hoc transformations to optimize cluster separability for downstream unsupervised tasks.

Future work will focus on addressing these limitations and extending the framework toward clinical applicability. A primary direction is the incorporation of clinically meaningful severity labels, such as pulmonary function test (PFT) measurements or expert-annotated severity scores, enabling supervised calibration of the regression output. In addition, patient-level stratification and cross-subject evaluation will be explored to assess robustness under real-world deployment scenarios. More expressive generative architectures, including β-VAEs or conditional VAEs, may be investigated to improve disentanglement and explicitly model pathological factors such as crackles and wheezes. Integrating attention mechanisms and temporal modeling could further enhance sensitivity to subtle acoustic variations across respiratory cycles. Additionally, clustering-oriented objectives, contrastive learning, or hybrid PCA+VAE approaches could be explored to improve latent space structure for unsupervised analyses. Finally, latent-space interpretability analysis and visualization will be pursued to link learned embeddings with clinically recognizable sound patterns, supporting explainable AI objectives in healthcare.

In conclusion, this work demonstrates that VAE-driven latent representation learning is a promising approach for respiratory sound analysis beyond traditional classification paradigms. By framing respiratory assessment as a continuous inference problem and leveraging generative modeling to structure the feature space, the proposed framework lays the groundwork for scalable, robust, and interpretable respiratory health monitoring systems. With improved labeling, validation, and latent-space interpretability—potentially guided by clustering or contrastive analyses—such models can support objective auscultation, assist clinical decision-making, and contribute meaningfully to next-generation digital respiratory diagnostics.

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